🏆 Infosys SE Elite Crack Vault — 12 College ITM Quantitative Aptitude 70Q Exam Compendium Elite

> **Target Role: Infosys Systems Engineer (SE)** (3.6 – 4.0 LPA Recruitment Drive)
> **Curation Standard: 100% Pure SE Role Specific** (All SP/DSE Upper-Level & Practice Variants Purged)
> **Total Curated Questions in Topic: 49 Questions**
> **Question Breakdown:** 🔴 `17 Must-Solve` | 🟡 `32 Core SE Level`
> **Verification Score:** 100% Mathematically & Deductively Proven with Zero Hallucination

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Question 1: Infosys SE Question 1 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
In how many ways can the letters of COMMITTEE be arranged so that no two vowels are adjacent?

Options:
- **(A)** 5400
- **(B)** 3600
- **(C)** 10800
- **(D)** 2700

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `5400`

Detailed Step-by-Step Solution & Mathematical Derivation:
Word COMMITTEE contains 9 letters: C(1), O(1), M(2), I(1), T(2), E(2).
Vowels: O, I, E, E (4 vowels, with two E's).
Consonants: C, M, M, T, T (5 consonants, with two M's and two T's).
1) Arrange the 5 consonants first:
Number of ways = 5! / (2! × 2!) = 120 / 4 = 30 ways.
2) 5 consonants create 6 available gaps: _ C _ M _ M _ T _ T _
To ensure no two vowels are adjacent, choose 4 gaps out of 6 for the 4 vowels:
C(6, 4) = 15 ways.
3) Arrange the 4 vowels (O, I, E, E) in the chosen 4 gaps:
Number of ways = 4! / 2! = 24 / 2 = 12 ways.
Total arrangements = 30 × 15 × 12 = 5,400.

Question 2: Infosys SE Question 2 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A committee of 5 is to be formed from 5 boys and 4 girls with at least 2 boys and at least 2 girls. In how many ways can this be done?

Options:
- **(A)** 90
- **(B)** 100
- **(C)** 105
- **(D)** 120

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (B)** — 100

Detailed Step-by-Step Solution & Mathematical Derivation:
Committee of 5 from 5 boys and 4 girls with at least 2 boys and at least 2 girls.
Possible compositions (Boys, Girls):
- Case 1: 2 Boys and 3 Girls => C(5, 2) × C(4, 3) = 10 × 4 = 40 ways.
- Case 2: 3 Boys and 2 Girls => C(5, 3) × C(4, 2) = 10 × 6 = 60 ways.
Total possible ways = 40 + 60 = 100 ways.

Question 3: Infosys SE Question 3 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
Two fair dice are thrown. Given that at least one die shows 4, what is the probability that the sum is a prime number?

Options:
- **(A)** 4/11
- **(B)** 1/3

- **(C)** 5/11
- **(D)** 4/9

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `4/11`

Detailed Step-by-Step Solution & Mathematical Derivation:
Two fair dice thrown. Total outcomes with at least one 4:
Sample space S = {(4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (1,4), (2,4), (3,4), (5,4), (6,4)} => n(S) = 11 outcomes.
Sums for each outcome:
- (4,1) -> 5 (prime)
- (4,2) -> 6
- (4,3) -> 7 (prime)
- (4,4) -> 8
- (4,5) -> 9
- (4,6) -> 10
- (1,4) -> 5 (prime)
- (2,4) -> 6
- (3,4) -> 7 (prime)
- (5,4) -> 9
- (6,4) -> 10

Favorable outcomes where sum is prime: {(4,1), (4,3), (1,4), (3,4)} => 4 outcomes.
Probability = 4 / 11.

Question 4: Infosys SE Question 4 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A bag has 5 red, 4 blue and 3 green balls. Three balls are drawn at random without replacement. What is the probability that all three are of different colours?

Options:
- **(A)** 3/11
- **(B)** 1/4
- **(C)** 2/11
- **(D)** 5/22

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `3/11`

Detailed Step-by-Step Solution & Mathematical Derivation:
Total balls = 5 red + 4 blue + 3 green = 12 balls.
3 balls drawn at random without replacement.
Total ways = C(12, 3) = (12 × 11 × 10) / (3 × 2 × 1) = 220.
Favorable ways to get one of each colour = C(5, 1) × C(4, 1) × C(3, 1) = 5 × 4 × 3 = 60.
Probability = 60 / 220 = 3 / 11.

Question 5: Infosys SE Question 5 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_01_page_02.png

Question Text:
A shopkeeper marks his goods 40% above cost price and allows two successive discounts of 10% and 15%. His net profit percentage is:

Options:
- **(A)** 7.1%
- **(B)** 5.5%
- **(C)** 8.6%
- **(D)** 6.4%

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `7.1%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Let Cost Price (CP) = 100.
Marked Price (MP) = 100 × 1.40 = 140.
Successive discounts of 10% and 15%:
Selling Price (SP) = 140 × (1 - 0.10) × (1 - 0.15) = 140 × 0.90 × 0.85 = 126 × 0.85 = 107.1.
Net profit = 107.1 - 100 = 7.1%.

Question 6: Infosys SE Question 6 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
A, B and C can individually finish a job in 12, 18 and 24 days respectively. They work on alternate days in the order A, B, C, A, B, C, ... (one person per day, starting with A). On which day is the job completed?

Options:
- **(A)** 16th
- **(B)** 17th
- **(C)** 18th
- **(D)** 19th

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (B)** — 17th

Detailed Step-by-Step Solution & Mathematical Derivation:
Let total work = LCM(12, 18, 24) = 72 units.
Daily efficiencies:
- A = 72 / 12 = 6 units/day
- B = 72 / 18 = 4 units/day
- C = 72 / 24 = 3 units/day
They work on alternate days: A on Day 1, B on Day 2, C on Day 3, A on Day 4, etc.
In 1 cycle of 3 days, work done = 6 + 4 + 3 = 13 units.
In 5 full cycles (15 days), work done = 5 × 13 = 65 units.
Remaining work = 72 - 65 = 7 units.
- Day 16: A works and completes 6 units. Total work = 65 + 6 = 71 units. Remaining = 1 unit.
- Day 17: B works. Since B does 4 units/day, B finishes the remaining 1 unit in 1/4 of a day.
Therefore, the job is completed on the 17th day.

Question 7: Infosys SE Question 7 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
Two trains of lengths 240 m and 160 m run on parallel tracks. Running in opposite directions they cross each other in 8 s; running in the same direction the faster overtakes the slower in 40 s. The speed of the faster train is:

Options:
- **(A)** 90 km/h
- **(B)** 108 km/h
- **(C)** 120 km/h
- **(D)** 72 km/h

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (B)** — `108 km/h`

Detailed Step-by-Step Solution & Mathematical Derivation:
Total distance to clear = 240 m + 160 m = 400 m.
Let speed of faster train = v1 m/s and slower train = v2 m/s.
1) Running in opposite directions, relative speed = v1 + v2 = 400 / 8 = 50 m/s.
2) Running in same direction, relative speed = v1 - v2 = 400 / 40 = 10 m/s.
Adding the two equations: 2·v1 = 60 => v1 = 30 m/s.
Converting to km/h: 30 × (18 / 5) = 108 km/h.

Question 8: Infosys SE Question 8 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
In an election, 20,000 voters were registered. 80% voted, and 5% of the votes cast were invalid. The winner got 55% of the valid votes. By how many votes did he win?

Options:
- **(A)** 1520
- **(B)** 1600
- **(C)** 1360
- **(D)** 1440

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `1520`

Detailed Step-by-Step Solution & Mathematical Derivation:
Total registered voters = 20,000.
Voters who voted = 80% of 20,000 = 16,000.
Invalid votes = 5% of 16,000 = 800.
Valid votes = 16,000 - 800 = 15,200.
Winner obtained 55% of valid votes => Loser obtained (100% - 55%) = 45%.
Winning margin = 55% - 45% = 10% of valid votes.
Winning margin = 0.10 × 15,200 = 1,520 votes.

Question 9: Infosys SE Question 9 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
The difference between compound interest (annual) and simple interest on a sum for 3 years at 10% p.a. is ₹310. Find the sum.

Options:
- **(A)** ₹8,000
- **(B)** ₹9,000
- **(C)** ₹10,000
- **(D)** ₹12,000

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (C)** — ₹10,000

Detailed Step-by-Step Solution & Mathematical Derivation:
Difference between CI and SI for 3 years at rate r%:
Formula: Difference = $P \times (r / 100)^2 \times (3 + r / 100)$.
Given r = 10% and Difference = ₹310:
$310 = P \times (0.10)^2 \times (3 + 0.10)$
$310 = P \times 0.01 \times 3.10 = P \times 0.031$
$P = 310 / 0.031 = ₹10,000$.

Question 10: Infosys SE Question 10 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 1 - Q25)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_01_page_03.png

Question Text:
₹12,000 is invested partly at 8% simple interest and the rest at 10% compounded annually. After 2 years the total interest is ₹2,270. How much was invested at simple interest?

Options:
- **(A)** ₹4,000
- **(B)** ₹5,000
- **(C)** ₹6,000
- **(D)** ₹7,000

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (B)** — `₹5,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
Let sum invested at 8% SI = ₹x.
Then sum invested at 10% CI = ₹(12,000 - x).
Time = 2 years.
- Simple Interest at 8% for 2 years = (x × 8 × 2) / 100 = 0.16x.
- Compound Interest at 10% for 2 years = (12,000 - x) × [(1 + 0.10)^2 - 1] = (12,000 - x) × 0.21 = 2,520 - 0.21x.
Total interest = 0.16x + 2,520 - 0.21x = 2,520 - 0.05x.
Given total interest = ₹2,270:
2,520 - 0.05x = 2,270
0.05x = 250
x = 250 / 0.05 = ₹5,000.

Question 11: Infosys SE Question 11 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
In how many ways can the letters of the word MANAGEMENT be arranged so that all the vowels come together?

Options:
- **(A)** 7560
- **(B)** 15120

- **(C)** 3780
- **(D)** 30240

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `7560`

Detailed Step-by-Step Solution & Mathematical Derivation:
MANAGEMENT has M2, A2, N2, G1, E2, T1 (10 letters). Vowels A,A,E,E form one block: internal arrangements = 4!/(2!2!) = 6. Treating the block as one unit alongside the 6 consonants (M2,N2,G1,T1) gives 7 units: 7!/(2!2!) = 1260. Total = 1260 × 6 = 7,560.

Question 12: Infosys SE Question 12 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
From a group of 6 men and 5 women, a team of 4 is to be selected such that it contains at most 2 women. In how many ways can this be done?

Options:
- **(A)** 265
- **(B)** 295
- **(C)** 310
- **(D)** 325

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `265`

Detailed Step-by-Step Solution & Mathematical Derivation:
0 women: C(6,4)=15. 1 woman: C(6,3)×C(5,1)=100. 2 women: C(6,2)×C(5,2)=150. Total = 15+100+150 = 265.

Question 13: Infosys SE Question 13 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
Three coins are tossed together, and it is known that at least one head appeared. What is the probability that exactly two heads appeared?

Options:
- **(A)** 3/7
- **(B)** 1/2
- **(C)** 3/8
- **(D)** 2/7

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `3/7`

Detailed Step-by-Step Solution & Mathematical Derivation:
Excluding TTT leaves 7 equally likely outcomes. Exactly two heads: HHT, HTH, THH = 3 outcomes. Probability = 3/7.

Question 14: Infosys SE Question 14 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A box contains 6 defective and 14 non-defective bulbs. Two bulbs are drawn at random without replacement. What is the probability that at least one is defective?

Options:
- **(A)** 99/190
- **(B)** 91/190
- **(C)** 33/95
- **(D)** 29/95

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `99/190`

Detailed Step-by-Step Solution & Mathematical Derivation:
P(no defective) = (14×13)/(20×19) = 91/190. P(at least one defective) = 1 − 91/190 = 99/190.

Question 15: Infosys SE Question 15 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A trader marks his goods at a price that would give him a 25% profit, but he actually gives a discount of 12% on the marked price and also uses a weight that is 10% less than it should be. Find his actual profit percentage.

Options:
- **(A)** 20%
- **(B)** 22.2%
- **(C)** 25%
- **(D)** 18%

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (B)** — 22.2%

Detailed Step-by-Step Solution & Mathematical Derivation:
Take CP = 100 per unit weight, so marked price = 125 (25% intended profit). After a 12% discount, SP = 125 × 0.88 = 110. But he delivers only 0.9 units (10% short weight) whose true cost is 90. Profit = 110 − 90 = 20 on a cost of 90, giving 20/90 × 100 ≈ 22.2%.

Question 16: Infosys SE Question 16 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE / HIGH RECURRENCE]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A and B together can complete a task in 15 days. They worked together for 5 days, after which A left and B completed the remaining work alone in 20 more days. In how many days can A alone complete the task?

Options:
- **(A)** 20 days
- **(B)** 24 days
- **(C)** 25 days
- **(D)** 30 days

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (D)** — `30 days`

Detailed Step-by-Step Solution & Mathematical Derivation:
In 5 days together, they finish 5/15 = 1/3. The remaining 2/3 is done by B alone in 20 days, so B's rate = (2/3)/20 = 1/30/day (B alone takes 30 days). Combined rate = 1/15, so A's rate = 1/15 − 1/30 = 1/30/day — A alone also takes 30 days.

Question 17: Infosys SE Question 17 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE / HIGH RECURRENCE]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)

(Infosys System Engineer ITM Mock Paper 2 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_02_page_02.png

Question Text:
A boat's speed in still water is 15 km/h. It travels 60 km downstream and returns upstream to the starting point, taking a total of 9 hours. Find the speed of the stream.

Options:
- **(A)** 4 km/h
- **(B)** 5 km/h
- **(C)** 6 km/h
- **(D)** 3 km/h

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (B)** — `5 km/h`

Detailed Step-by-Step Solution & Mathematical Derivation:
60/(15+s) + 60/(15−s) = 9. Simplifying: 1800 = 9(225 − s^2), so s^2 = 25, s = 5 km/h.

Question 18: Infosys SE Question 18 🟡 `[⭐ CORE SE LEVEL | HIGH PROBABILITY]`

> 🎯 **Exam Priority:** 🟡 `[⭐ CORE SE LEVEL]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level`

> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 2 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_02_page_03.png

Question Text:
A town's population increases by 10% in the first year and decreases by 8% in the second year. If the population at the end of two years is 27,324, what was the population at the start?

Options:
- **(A)** 27,000
- **(B)** 26,500
- **(C)** 26,000
- **(D)** 27,500

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `27,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
P × 1.10 × 0.92 = 27,324, so P × 1.012 = 27,324, giving P = 27,000.

Question 19: Infosys SE Question 19 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 2 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_02_page_03.png

Question Text:
A sum of money amounts to ₹13,000 after 3 years and to ₹15,000 after 5 years at the same rate of simple interest. Find the rate of interest per annum.

Options:
- **(A)** 8%
- **(B)** 10%
- **(C)** 12%
- **(D)** 9%

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (B)** — `10%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Interest for 2 years (year 5 − year 3) = 15,000 − 13,000 = 2,000, so 1 year's interest = 1,000. Principal = 13,000 − 3×1,000 = 10,000. Rate = 1,000/10,000 × 100 = 10%.

Question 20: Infosys SE Question 20 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 2 - Q25)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_02_page_03.png

Question Text:
₹16,000 is invested for 2 years at compound interest, compounded annually, and yields the same interest as ₹18,000 invested for 2 years at 10% simple interest. Find the rate of compound interest (approximately).

Options:
- **(A)** 9.5%
- **(B)** 10.5%
- **(C)** 11.5%
- **(D)** 12.5%

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (B)** — 10.5%

Detailed Step-by-Step Solution & Mathematical Derivation:
SI on ₹18,000 at 10% for 2 years = ₹3,600. Setting CI on ₹16,000 equal: $16,000\times((1+r)^2 - 1) = 3,600$ gives $(1+r)^2 = 1.225$, so $r \approx 10.5\%$.

Question 21: Infosys SE Question 21 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
In how many ways can 8 different books be distributed among 3 students so that one student gets 4 books and the other two get 2 books each?

Options:
- **(A)** 1260
- **(B)** 2520
- **(C)** 630
- **(D)** 420

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `1260`

Detailed Step-by-Step Solution & Mathematical Derivation:
Choose which of the 3 students gets 4 books: 3 ways. Choose that student's 4 books: C(8,4)=70. Split the remaining 4 books between the other two specific students, 2 each: C(4,2)=6. Total = 3×70×6 = 1,260.

Question 22: Infosys SE Question 22 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A committee of 6 is to be formed from 7 men and 6 women such that the committee has more men than women. In how many ways can this be done?

Options:
- **(A)** 658
- **(B)** 686
- **(C)** 700
- **(D)** 623

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `658`

Detailed Step-by-Step Solution & Mathematical Derivation:
Valid (men,women) splits with men>women, total=6: (6,0): C(7,6)×C(6,0)=7. (5,1): C(7,5)×C(6,1)=21×6=126. (4,2): C(7,4)×C(6,2)=35×15=525. Total = 7+126+525 = 658. (Cross-check: adding the equal split (3,3)=700 and women>men splits=358 gives 658+700+358=1,716=C(13,6), confirming the count.)

Question 23: Infosys SE Question 23 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
Bag A has 4 red and 6 black balls; Bag B has 7 red and 3 black balls. A bag is chosen at random and one ball is drawn from it. Given that the ball drawn is red, what is the probability that it came from Bag A?

Options:
- **(A)** 4/11
- **(B)** 7/11
- **(C)** 2/5
- **(D)** 1/2

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — 4/11

Detailed Step-by-Step Solution & Mathematical Derivation:
P(Red|A)=4/10, P(Red|B)=7/10, each bag chosen with probability 1/2. P(Red) = ½×0.4 + ½×0.7 = 0.55. By Bayes' theorem, P(A|Red) = (½×0.4)/0.55 = 0.2/0.55 = 4/11.

Question 24: Infosys SE Question 24 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_03_page_02.png

Question Text:
A drawer has 5 red pens, 4 blue pens and 3 black pens. Four pens are drawn at random. What is the probability that exactly 2 are red and 2 are blue?

Options:
- **(A)** 4/33
- **(B)** 3/33
- **(C)** 5/33
- **(D)** 4/29

Sokware effort estimation has resisted four decades of methodological refinement, and the reason is not, as oken supposed, a lack of adequate techniques. Function point analysis, story point velocity tracking and Monte Carlo simulation are all, in controlled retrospective studies, reasonably well calibrated. The persistent gap between estimated and actual delivery dates arises instead from a structural asymmetry: the costs of underestimating are diffuse and delayed, distributed across a project's eventual overrun, whereas the costs of overestimating are immediate and attributable, since a padded estimate can lose a bid or provoke management scrutiny before the work even begins. Estimators therefore face an incentive gradient that has nothing to do with statistical accuracy and everything to do with organisational consequence, and this gradient reliably pushes estimates toward optimism regardless of how sophisticated the underlying model is. Attempts to correct this by adding explicit contingency buffers frequently fail for a related reason: once a buffer becomes visible in a schedule, it is treated by stakeholders as slack to be reclaimed rather than risk to be reserved, a phenomenon sometimes called the student syndrome, in reference to the tendency to begin a task only as its deadline approaches.

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `4/33`

Detailed Step-by-Step Solution & Mathematical Derivation:
Favourable = C(5,2)×C(4,2) = 10×6 = 60. Total ways = C(12,4) = 495. Probability = 60/495 = 4/33.

Question 25: Infosys SE Question 25 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
A dealer's cost price for a mixture of two rice varieties (mixed 3:2 by weight at ₹40/kg and ₹60/kg) works out to ₹48/kg. He marks up the price by 25% and sells it, but uses a false weight that gives only 900 g for every 1 kg paid for. Find his actual profit percentage.

Options:
- **(A)** 38.9%
- **(B)** 35%
- **(C)** 41.7%
- **(D)** 33.3%

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `38.9%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Marked-up SP for a claimed 1 kg = 48×1.25 = ₹60. True cost of the 900 g actually given = 0.9×48 = ₹43.2. Profit = 60−43.2 = ₹16.8 on a true cost of ₹43.2, giving 16.8/43.2×100 ≈ 38.9%.

Question 26: Infosys SE Question 26 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
Pipe A can fill a tank in 12 hours, Pipe B in 15 hours, and a leak can empty the full tank in 20 hours. All three are opened together for 4 hours, after which the leak is plugged. How much more time do A and B together need to fill the remaining tank?

Options:
- **(A)** 4 hours
- **(B)** 3 hours
- **(C)** 5 hours
- **(D)** 6 hours

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `4 hours`

Detailed Step-by-Step Solution & Mathematical Derivation:
Combined rate (A+B−leak) = 1/12+1/15−1/20 = 5/60+4/60−3/60 = 1/10 per hour. In 4 hours, 4/10 = 2/5 of the tank fills, leaving 3/5. A+B's combined rate (no leak) = 1/12+1/15 = 3/20 per hour. Time for remaining 3/5 = (3/5)÷(3/20) = 4 hours.

Question 27: Infosys SE Question 27 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
In a 1000 m race, A beats B by 100 m, and in a separate 1000 m race, B beats C by 100 m (each runner

maintains a constant speed across races). By how many metres does A beat C in a 1000 m race between them?

Options:
- **(A)** 190 m
- **(B)** 200 m
- **(C)** 180 m
- **(D)** 210 m

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `190 m`

Detailed Step-by-Step Solution & Mathematical Derivation:
Speed ratio A:B = 1000:900 = 10:9 (A beats B by 100 m in 1000 m). Similarly B:C = 10:9. So A:C = (10/9)×(10/9) = 100:81. When A covers 1000 m, C covers (81/100)×1000 = 810 m. A beats C by 1000−810 = 190 m.

Question 28: Infosys SE Question 28 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
An alloy contains 60% copper and 40% zinc. 20 kg of this alloy is melted and mixed with x kg of pure copper to form a new alloy containing 75% copper. Find x.

Options:
- **(A)** 12 kg
- **(B)** 10 kg
- **(C)** 15 kg
- **(D)** 8 kg

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `12 kg`

Detailed Step-by-Step Solution & Mathematical Derivation:
Copper in original alloy = 0.6×20 = 12 kg; zinc = 8 kg (unchanged after adding pure copper). For 75% copper (25% zinc): 8 = 0.25×(20+x), so 20+x = 32, x = 12 kg.

Question 29: Infosys SE Question 29 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
₹8,000 is invested for 1 year at 10% per annum, compounded half-yearly. Find the difference between the compound interest and the simple interest earned on the same sum, at the same rate, for the same period.

Options:
- **(A)** ₹20
- **(B)** ₹40
- **(C)** ₹80
- **(D)** ₹100

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — ₹20

Detailed Step-by-Step Solution & Mathematical Derivation:
Half-yearly compounding: 2 periods at 5% each. Amount = 8000×1.05×1.05 = 8,820, so CI = ₹820. SI = 8000×10%×1 = ₹800. Difference = 820−800 = ₹20.

Question 30: Infosys SE Question 30 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 3 - Q25)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_03_page_03.png

Question Text:
In what time will ₹8,000 amount to ₹9,261 at 5% per annum, compounded annually?

Options:
- **(A)** 2 years
- **(B)** 3 years
- **(C)** 4 years
- **(D)** 3.5 years

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (B)** — `3 years`

Detailed Step-by-Step Solution & Mathematical Derivation:
9,261/8,000 = 1.157625 = 1.05^3 (since 1.05^2=1.1025, ×1.05=1.157625). So the time is 3 years.

Question 31: Infosys SE Question 31 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
In how many ways can 6 men and 4 women be seated at a round table such that no two women sit together?

Options:
- **(A)** 43,200
- **(B)** 86,400
- **(C)** 21,600
- **(D)** 14,400

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `43,200`

Detailed Step-by-Step Solution & Mathematical Derivation:
Seat 6 men in a circle: (6−1)! = 120 ways. This creates 6 gaps; choose and arrange 4 women in 4 of these 6 distinct gaps: P(6,4) = 6×5×4×3 = 360. Total = 120×360 = 43,200.

Question 32: Infosys SE Question 32 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_04_page_02.png

Question Text:
In how many ways can the letters of the word ORANGE be arranged so that the vowels occupy only the odd positions?

Options:
- **(A)** 36
- **(B)** 72
- **(C)** 18
- **(D)** 144

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `36`

Detailed Step-by-Step Solution & Mathematical Derivation:
ORANGE has 3 vowels (O,A,E) and 3 consonants (R,N,G), matching the 3 odd and 3 even positions exactly. Vowels in odd positions: 3! = 6. Consonants in even positions: 3! = 6. Total = 6×6 = 36.

Question 33: Infosys SE Question 33 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
Three machines A, B and C produce 25%, 35% and 40% of a factory's output respectively, with defect rates of 5%, 4% and 2%. An item is picked at random and found defective. What is the probability it came from machine C?

Options:
- **(A)** 16/69
- **(B)** 8/69
- **(C)** 23/69
- **(D)** 25/69

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `16/69`

Detailed Step-by-Step Solution & Mathematical Derivation:
P(D) = 0.25×0.05 + 0.35×0.04 + 0.40×0.02 = 0.0125+0.014+0.008 = 0.0345. P(C|D) = (0.40×0.02)/0.0345 = 0.008/0.0345 = 16/69.

Question 34: Infosys SE Question 34 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
5 boys and 3 girls are seated randomly in a row of 8 chairs. What is the probability that all 3 girls sit together?

Options:
- **(A)** 3/28
- **(B)** 1/8
- **(C)** 3/56
- **(D)** 5/28

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — 3/28

Detailed Step-by-Step Solution & Mathematical Derivation:
Treating the 3 girls as one block: favourable arrangements = 6!×3! = 720×6 = 4,320. Total arrangements = 8! = 40,320. Probability = 4320/40320 = 3/28.

Question 35: Infosys SE Question 35 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
A trader buys 80 kg of rice at ₹25/kg. He sells 50 kg at a 20% profit and wants his overall profit on all 80 kg to be 15%. At what price per kg must he sell the remaining 30 kg?

Options:
- **(A)** ₹26.67
- **(B)** ₹25
- **(C)** ₹28
- **(D)** ₹24

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `₹26.67`

Detailed Step-by-Step Solution & Mathematical Derivation:
Total CP = 80×25 = ₹2,000; target total SP (15% profit) = ₹2,300. SP from the first 50 kg (cost ₹1,250, at 20% profit) = ₹1,500. Remaining SP needed = 2300−1500 = ₹800 for 30 kg → ₹800/30 ≈ ₹26.67/kg.

Question 36: Infosys SE Question 36 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
A, B and C can complete a work alone in 20, 30 and 60 days respectively. All three start together. After 4 days, B leaves; after 3 more days, C also leaves, and A finishes the rest alone. In how many total days is the work completed?

Options:
- **(A)** 15 days
- **(B)** 14 days
- **(C)** 16 days
- **(D)** 13 days

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `15 days`

Detailed Step-by-Step Solution & Mathematical Derivation:
Rates (LCM 60): A=3/60, B=2/60, C=1/60 per day. First 4 days (all three): 4×6/60=24/60 done. Next 3 days (A+C only): 3×4/60=12/60 done — total 36/60=3/5 done, 2/5=24/60 remains. A alone at 3/60/day takes 24/60÷3/60 = 8 more days. Total = 4+3+8 = 15 days.

Question 37: Infosys SE Question 37 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
Two trains, 300 km apart, move towards each other on the same track. Train A departs at 8:00 AM at 60 km/h; Train B departs at 8:30 AM at 90 km/h. At what time do they meet?

Options:
- **(A)** 10:18 AM
- **(B)** 10:12 AM
- **(C)** 10:06 AM
- **(D)** 10:24 AM

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `10:18 AM`

Detailed Step-by-Step Solution & Mathematical Derivation:
From 8:00 to 8:30, only A moves: 60×0.5=30 km, leaving 270 km. From 8:30, combined speed = 150 km/h; time = 270/150 = 1.8 hr = 1 hr 48 min. Meeting time = 8:30 AM + 1:48 = 10:18 AM.

Question 38: Infosys SE Question 38 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
In a two-candidate election, the winner received 60% of votes cast and won by 4,800 votes. If 20% of registered voters did not vote and there were no invalid votes, how many voters were registered?

Options:
- **(A)** 30,000
- **(B)** 28,000
- **(C)** 32,000
- **(D)** 24,000

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `30,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
Winner's margin (60%−40%=20% of votes cast) = 4,800 → votes cast = 4800/0.20 = 24,000. Since this is 80% of registered voters: registered = 24000/0.80 = 30,000.

Question 39: Infosys SE Question 39 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
Find the compound interest on ₹12,000 for 1.5 years at 10% per annum, compounded annually (interest for the fractional half-year is simple interest on the amount at the end of the first year).

Options:
- **(A)** ₹3,860
- **(B)** ₹3,800
- **(C)** ₹3,960
- **(D)** ₹3,720

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — ₹3,860

Detailed Step-by-Step Solution & Mathematical Derivation:
Year 1: 12000×1.10 = 13,200 (CI so far = 1,200). For the next half-year, SI on 13,200 at 10% for 0.5 yr = 660. Total amount = 13,860; total CI = 13860−12000 = ₹3,860.

Question 40: Infosys SE Question 40 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 4 - Q25)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_04_page_03.png

Question Text:
The compound interest on a sum for 2 years is ₹1,050, and for 3 years is ₹1,655, at the same annual rate. The simple interest on the same sum for 2 years, at the same rate, is ₹1,000. Find the rate of interest per annum.

Options:
- **(A)** 10%
- **(B)** 12%
- **(C)** 8%
- **(D)** 15%

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `10%`

Detailed Step-by-Step Solution & Mathematical Derivation:
3rd-year interest = 1655−1050 = ₹605 — but more directly: $CI_2 - SI_2 = Pr^2 = 1050 - 1000 = 50$, and $SI_2 = 2Pr = 1000$ so Pr=500. Dividing, r = 50/500 = 0.10 = 10% (and P=₹5,000, verified: $5000 \times 1.1^3 = 6,655$, $CI_3 = 1,655$ ✓).

Question 41: Infosys SE Question 41 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q16)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 2
Visual Reference Plate: paper_05_page_02.png

Question Text:
A committee of 5 is to be formed from 6 men and 5 women. A specific man X must be included, a specific woman Y must be excluded, and the committee must include at least 1 woman. In how many ways can this be done?

Options:
- **(A)** 121
- **(B)** 126
- **(C)** 120
- **(D)** 115

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `121`

Detailed Step-by-Step Solution & Mathematical Derivation:
With X fixed in and Y fixed out, the remaining pool is 5 men + 4 women = 9 people, from which 4 more members are chosen: C(9,4)=126. Subtract the invalid case where all 4 additional members are men (no woman at all): C(5,4)=5. Valid = 126−5 = 121.

Question 42: Infosys SE Question 42 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q17)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
In how many ways can the 8 distinct letters of COMPUTER be arranged such that C and T have exactly 2 letters between them (in either order)?

Options:
- **(A)** 7200
- **(B)** 6000
- **(C)** 8400
- **(D)** 5040

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `7200`

Detailed Step-by-Step Solution & Mathematical Derivation:
Valid position-pairs with C and T exactly 3 apart: 5 pairs with C before T, plus 5 with T before C = 10 total. The remaining 6 distinct letters fill the remaining 6 positions in 6!=720 ways. Total = 10×720 = 7,200.

Question 43: Infosys SE Question 43 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q18)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
Two tickets are drawn one after another, without replacement, from a box containing tickets numbered 1 to 10. What is the probability that the number on the second ticket is greater than the number on the first?

Options:
- **(A)** 1/2
- **(B)** 9/20
- **(C)** 11/20
- **(D)** 2/5

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `1/2`

Detailed Step-by-Step Solution & Mathematical Derivation:
By symmetry, for any two distinct numbers drawn in some order, exactly half the time the second exceeds the first. Probability = 1/2 (confirmed via direct count: 45 favourable ordered outcomes out of 90).

Question 44: Infosys SE Question 44 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q19)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A mini-deck has 16 cards: 4 suits with 4 cards each. 4 cards are drawn at random. What is the probability that at least 2 of them share the same suit?

Options:
- **(A)** 391/455
- **(B)** 64/455
- **(C)** 327/455
- **(D)** 201/455

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `391/455`

Detailed Step-by-Step Solution & Mathematical Derivation:
P(all 4 different suits) = (4×4×4×4)/C(16,4) = 256/1820 = 64/455. P(at least 2 same suit) = 1−64/455 = 391/455.

Question 45: Infosys SE Question 45 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q20)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A shopkeeper runs a "Buy 3, Get 1 Free" offer on an item marked at ₹200 (costing him ₹120 each). A customer buys 4 items under this offer, paying for 3. What is the shopkeeper's profit percentage, based on his cost for all 4 items given?

Options:
- **(A)** 25%
- **(B)** 20%
- **(C)** 30%
- **(D)** 15%

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `25%`

Detailed Step-by-Step Solution & Mathematical Derivation:
Cost for 4 items = 4×120 = ₹480. Revenue (customer pays for 3 at ₹200) = ₹600. Profit = ₹120 on a cost of ₹480, giving 120/480×100 = 25%.

Question 46: Infosys SE Question 46 🟡 [☆ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [☆ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q21)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A, B and C can individually complete a work in 15, 20 and 30 days. They work together, and the total payment of ₹9,000 is distributed in proportion to each person's work contribution. How much should C be paid?

Options:
- **(A)** ₹2,000
- **(B)** ₹1,800
- **(C)** ₹2,200
- **(D)** ₹2,500

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `₹2,000`

Detailed Step-by-Step Solution & Mathematical Derivation:
Rates in the ratio 1/15:1/20:1/30 = 4:3:2 (over LCM 60). C's share = (2/9)×9000 = ₹2,000.

Question 47: Infosys SE Question 47 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q22)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
Two runners start from the same point on a 600 m circular track at the same time, in opposite directions, at 8 m/s and 10 m/s. After how many seconds will they both simultaneously be back at the starting point for the first time?

Options:
- **(A)** 300 s
- **(B)** 200 s
- **(C)** 150 s
- **(D)** 600 s

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — 300 s

Detailed Step-by-Step Solution & Mathematical Derivation:
A takes 600/8=75 s/lap; B takes 600/10=60 s/lap. Both are back at the start simultaneously at LCM(75,60) = 300 s.

Question 48: Infosys SE Question 48 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** Infosys SE Standard Level
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q23)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
A container has 80 litres of pure milk. 20 litres of the mixture is removed and replaced with water; this process is repeated once more. How much pure milk remains?

Options:
- **(A)** 45 litres
- **(B)** 50 litres
- **(C)** 40 litres
- **(D)** 48 litres

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `45 litres`

Detailed Step-by-Step Solution & Mathematical Derivation:
Using the repeated-dilution formula: Final = $80\times(1-20/80)^2 = 80\times(3/4)^2 = 80\times9/16 = 45$ litres.

Question 49: Infosys SE Question 49 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL]
> 📊 **Target Difficulty:** `Infosys SE Standard Level`
> 💡 **Preparation Advice:** Essential syllabus concept — highly recommended for sectional cutoff.

(Infosys System Engineer ITM Mock Paper 5 - Q24)
Section: Section 2: Technical Ability (Quantitative) | **Page:** 3
Visual Reference Plate: paper_05_page_03.png

Question Text:
At a certain rate of annual compound interest, a sum doubles itself in 5 years. In how many years will it become 8 times itself, at the same rate?

Options:
- **(A)** 15 years
- **(B)** 20 years
- **(C)** 10 years
- **(D)** 25 years

Correct Answer

> [!IMPORTANT]
> **Verified Answer Key**: (A)** — `15 years`

Detailed Step-by-Step Solution & Mathematical Derivation:
$(1+r)^5=2$ (doubling). For 8×: $(1+r)^n=8=2^3=[(1+r)^5]^3=(1+r)^{15}$, so n=15 years.
